Coupled Registration and Change Detection for Remote Sensing Images: A Joint Optimization Framework

Jiaxin Lu, Liangzhi Li, Tingyu Zhang, Gaoxiang Zhou, Kyle Gao, Zongfan Bai, and Zongyi Liu

***Abstract*—This article addresses a fundamental limitation in remote sensing image analysis: the error propagation caused by treating registration and change detection as independent sequential tasks. We propose a novel end-to-end joint learning framework that establishes bidirectional coupling between these two tasks, enabling mutual enhancement through shared optimization. The key insight is that registration quality directly impacts detection accuracy, while change detection feedback can guide more semantically meaningful alignment. To realize this joint optimization, we design a unified architecture where a contrastive learning-based module projects multitemporal images into a modality-invariant feature space, a graph-based regression module exploits spatial relationship priors for robust displacement prediction, and a generative adversarial detection module with global attention captures long-range semantic dependencies. Crucially, these components are connected through a multitask loss function that simultaneously optimizes structural consistency for registration and discriminative feature learning for detection, breaking the traditional pipeline's error accumulation. Experiments on four benchmark datasets demonstrate that this joint optimization paradigm significantly outperforms both traditional sequential approaches and state-of-the-art individual methods, achieving $F1$-scores of 93.10%, 90.60%, 95.46%, and 89.65% on WHU-CD, CDD, LEVIR-CD+, and Greenhouse Change Detection dataset (GHCD), respectively, validating that synergistic learning between registration and detection yields superior performance for handling geometric distortions and multitemporal heterogeneity.**

***Index Terms*—Attention mechanism, change detection, deep learning, image registration, remote sensing.**

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I. Introduction

With the rapid advancement of satellite remote sensing, the acquisition of high-resolution, multispectral, and multiplatform data has greatly facilitated quantitative remote sensing and global change studies [1], [2]. Change detection, which analyzes spatiotemporal dynamics of land surface, plays a critical role in environmental monitoring, disaster assessment, and urban planning [3], [4], [5].

In practice, multitemporal images inevitably suffer from geometric misalignment due to sensor viewing angles, platform errors, terrain, and atmospheric effects [6], [7]. Even subpixel misregistration can introduce substantial pseudo-changes that severely degrade detection accuracy [8], [9]. Thus, image registration has traditionally been performed as a prerequisite for change detection. However, this sequential pipeline treats the two tasks independently, which inherently limits overall performance [10]. The major drawback of the sequential paradigm is its unidirectional information flow: registration errors propagate to detection, while detection results, which contain valuable semantic cues, are not leveraged to refine alignment [11], [12]. In fact, unchanged regions can serve as reliable appearance anchors for registration, whereas changed areas should be excluded specifically from appearance-based similarity terms so that genuinely different objects are not forced to look alike [13], [14]. This exclusion does not imply that geometric displacement supervision or spatial regularization must be removed in changed regions: the sensor-induced coordinate transform remains defined over the image domain and is independent of whether the land cover content has changed. This distinction between appearance fidelity and geometric field regularity is central to the joint formulation below. This bidirectional dependency suggests that registration and change detection are coupled processes that can mutually benefit from joint optimization.

Moreover, registration methods optimized independently often adopt task-agnostic similarity metrics (e.g., cross correlation) that treat all pixels equally, potentially introducing biases in regions critical for change detection [15]. This issue is particularly acute for multimodal data, where radiometric differences further complicate correspondence establishment [16], [17]. To overcome these limitations, we propose an end-to-end joint learning framework that integrates registration and change detection through bidirectional coupling. As illustrated in Fig. 1, our framework formulates the two tasks as complementary subtasks within a multitask learning

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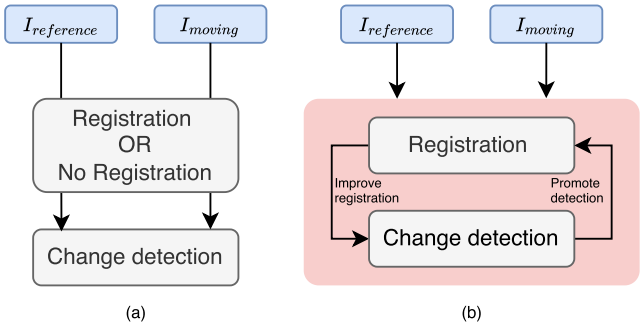


Fig. 1. Framework comparison. (a) Traditional sequential approach. (b) Proposed joint optimization framework.

paradigm, enabling shared feature representations and joint loss optimization. This design eliminates error accumulation, allows registration to adapt to detection feedback, and provides the detection module with registration-aware features.

Specifically, the registration module employs a spatial transformer network (STN) [18] integrated with graph convolutional networks (GCNs) to predict dense displacement fields, learning both local geometric relationships and global structural constraints. The change detection module adopts a conditional generative adversarial network (CGAN) [13] with a vision-transformer-enhanced encoder–decoder to capture long-range semantic dependencies, producing change maps with sharp boundaries. To achieve mutual enhancement, we design a multitask joint loss function with deep supervision: the registration loss incorporates structural similarity to guide alignment toward the detection objective, while the detection loss uses attention-weighted cross-entropy to emphasize regions where registration quality critically affects accuracy. Gradients from the detection loss backpropagate to refine registration, and improved alignment in turn boosts change discrimination, forming a virtuous cycle.

In summary, the main contributions are: 1) an end-to-end joint learning framework that couples registration and change detection via bidirectional optimization; 2) a registration module combining STN and GCN for adaptive, nonlinear alignment optimized for change detection; 3) integration of vision transformers into a CGAN-based detection module to capture both local details and global context; and 4) a multitask joint loss with deep supervision that eliminates error accumulation. Extensive experiments on benchmark datasets demonstrate superior performance in detection accuracy and robustness to geometric distortions over existing methods.

II. Related Works

Remote sensing image registration, as a fundamental prerequisite for change detection and other multitemporal analysis tasks, has evolved significantly over the past decades. Traditional registration methods primarily relied on handcrafted features such as SIFT, SURF, and their variants to establish correspondences between images. These approaches typically followed a sequential pipeline of feature detection, description, matching, and transformation estimation. While effective in controlled scenarios, these methods often struggled with the complex geometric distortions, radiometric variations, and temporal changes inherent in remote sensing imagery, particularly when dealing with multimodal data or large viewpoint differences. The emergence of deep learning has revolutionized the field of remote sensing image registration by enabling automatic feature learning and end-to-end optimization [15], [19]. Early deep learning approaches focused on enhancing individual components of the traditional pipeline, such as developing more robust feature descriptors or improving matching strategies. For instance, researchers proposed Siamese network architectures to learn discriminative feature representations specifically tailored for remote sensing image matching [20], [21]. These networks demonstrated superior performance in handling variations in illumination, scale, and perspective compared to conventional handcrafted features.

As the field progressed, attention shifted toward developing more integrated approaches that could learn the entire registration process in an end-to-end manner. The introduction of STNs marked a significant advancement by enabling the direct prediction of geometric transformations without explicit feature matching [18]. These networks could learn to estimate complex spatial transformations through backpropagation, making them particularly suitable for handling nonrigid deformations commonly encountered in remote sensing applications. Building upon this foundation, researchers incorporated attention mechanisms and multiscale processing strategies to enhance the network's ability to capture both local and global correspondence patterns [22], [23], [24]. Recent years have witnessed the emergence of semantic-guided registration methods that leverage high-level understanding to improve alignment accuracy [25]. These approaches integrate semantic segmentation networks with registration frameworks to ensure consistency at both the pixel and semantic levels. The incorporation of semantic information has proven particularly beneficial when dealing with images acquired under different conditions or from different sensors, as it helps establish reliable correspondences based on structural and contextual similarities rather than purely radiometric features.

The advent of transformer architectures has introduced new capabilities in modeling long-range dependencies and handling complex geometric relationships [26], [27]. Vision transformers adapted for registration tasks have demonstrated remarkable success in establishing accurate correspondences across different imaging conditions and modalities [28], [29]. The self-attention mechanism inherent in these architectures enables the capture of global contextual information and the modeling of sophisticated spatial relationships, leading to more robust registration results. Some researchers have further enhanced these approaches by incorporating hierarchical feature processing and multihead attention mechanisms to handle the varying scales and complexities present in remote sensing imagery [30]. Parallel to these developments, similarity-based registration methods have evolved to directly learn adaptive similarity metrics between image pairs [31], [32]. These approaches utilize deep neural networks to compute context-aware similarity measures that are more robust than traditional correlation-based metrics. The integration of multiscale processing and progressive refinement strategies has enabled these methods to handle large displacement fields and

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complex geometric transformations effectively [33]. Despite these advances, challenges remain in achieving robust and efficient registration for operational remote sensing applications. Current research trends focus on developing more lightweight architectures that can process high-resolution imagery efficiently, improving the interpretability of learned transformations, and addressing the scalability issues associated with large-scale applications [34], [35]. Additionally, there is growing interest in unsupervised and weakly supervised registration methods that can reduce the dependency on manually annotated training data while maintaining high accuracy [36], [37].

Change detection methodologies have undergone substantial evolution with the advancement of deep learning techniques [38]. Traditional methods based on spectral features measured intertemporal similarity through differencing, ratioing, and vector analysis to extract change regions [10]. While computationally efficient, these approaches struggled to fully exploit the spatiotemporal information in large remote sensing datasets and lacked sensitivity to local detail changes and small targets [39]. Deep learning-based change detection methods can be categorized into direct methods and differencing methods based on how temporal images are integrated. Direct methods input paired temporal images into deep neural networks, extracting change information through hierarchical convolution and pooling operations. Representative works include Siamese network structures that process pre- and post-change images separately before generating multiscale difference features through fusion modules [40]. Researchers have embedded spatial attention modules in encoders to enhance local region weights and introduced contrast loss functions to mitigate class imbalance [41]. Pyramid pooling has been employed to capture semantic information from different receptive fields, with self-attention mechanisms implementing long-range dependency modeling [23]. Direct methods fully utilize spatiotemporal context from original images to learn discriminative change features, though they typically incur higher computational complexity and memory consumption.

Differencing methods first perform algebraic operations on temporal image pairs to obtain difference images highlighting change regions, then classify these into change and nonchange categories using traditional classifiers or convolutional neural networks. Techniques for constructing difference images include pixel differencing [11], log-ratio methods, and change vector analysis [42], which partially suppress imaging differences and radiometric noise. Researchers have further optimized difference images through dual-attention-based enhancement modules in encoding stages [43], generative adversarial networks for correcting pseudo-changes from radiometric distortions [13], and joint probabilistic graphical models for error correction in difference maps [44]. Comparative studies have examined performance differences among early fusion, late fusion, and cross-layer fusion frameworks [45]. While differencing methods offer more concise detection processes and computational efficiency, the limited expressiveness of difference features leads to loss of detail information and poor adaptability to complex scenes.

Recent advances have explored the integration of emerging artificial intelligence technologies with convolutional neural networks to enhance change detection performance. Attention mechanisms explicitly model correlations between features or regions, enabling networks to focus on significantly changed channels and locations, partially alleviating semantic gap problems [12], [46], [47]. Graph neural networks utilize graph structures to describe topological relationships within images, capable of mining high-order information at superpixel or object levels, producing detection results with more refined and complete boundaries [48], [49]. Vision transformers employ self-attention to establish long-range dependencies, overcoming the limited receptive field constraints of convolutional neural networks and demonstrating advantages in capturing global change patterns [28], [50]. These technical innovations have opened broad avenues for advancing change detection methodologies. However, challenges persist when processing high-resolution, large-scale remote sensing images with nonrigid deformations, as most existing methods lack explicit mechanisms to handle geometric misalignment and focus predominantly on feature fusion in encoding stages while paying insufficient attention to detail restoration in decoding stages.

The recognition of interdependencies between registration and change detection has motivated recent exploration of joint optimization strategies. The continuous evolution of deep learning-based methods has significantly improved the reliability and accuracy of both registration and change detection individually [51]. These advancements have enabled new applications in environmental monitoring and multitemporal analysis [52]. The integration of emerging technologies such as graph neural networks and adaptive attention mechanisms promises to further advance the field, particularly in handling challenging scenarios involving multitemporal remote sensing data. Related studies have shown that unreliable correspondences should be suppressed in the appearance-matching pathway: an affinity-derived change prior can down-weight change pixels in multimodal image translation [13], feature-metric alignment can use the estimated change probability as a reliability weight [14], and learnable occlusion masks can filter locations that have no valid correspondence during feature warping [53]. These findings motivate a change-excluded appearance term, while leaving transformation-field regularization conceptually separate. However, to date, limited research has systematically investigated the potential benefits of coupling registration and change detection within a unified learning framework. This gap motivates our proposed approach, which establishes explicit bidirectional connections between these two tasks to achieve collaborative optimization and superior overall performance.

III. Methodology

A. Framework Overview

The proposed framework addresses the fundamental challenge of jointly optimizing image registration and change detection for cross-modal remote sensing imagery, as illustrated in Fig. 2. Given a pair of cross-modal images (I_x, I_y),

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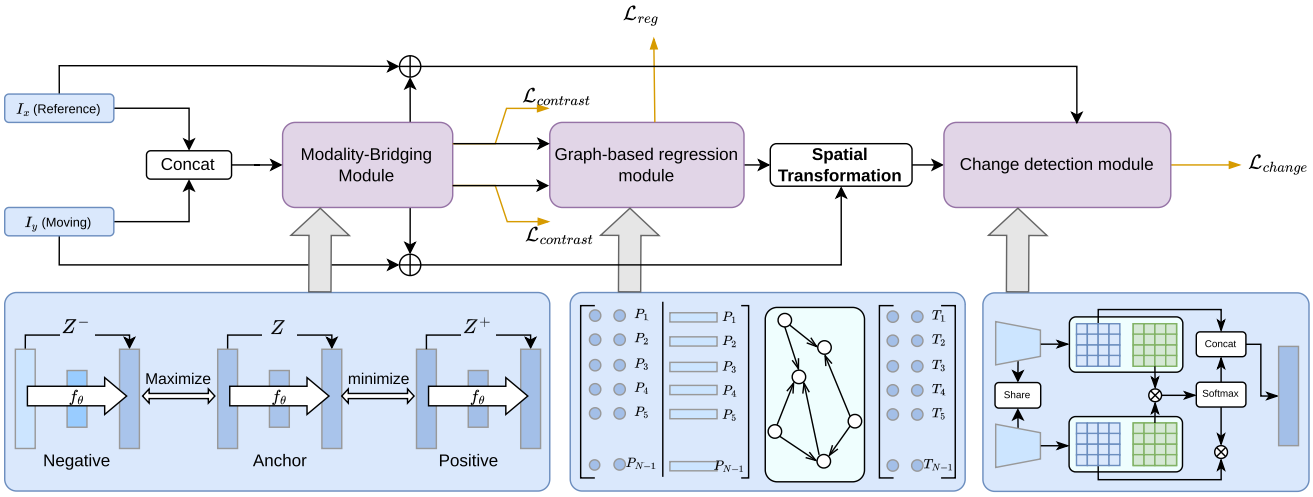


Fig. 2. Overall architecture of the proposed framework with three main components: similarity mapping, matchpoint regression, and change detection modules.

where I_x denotes the reference image and I_y denotes the sensed image, the framework establishes a bidirectional information flow that enables mutual enhancement between geometric alignment and semantic change extraction.

The architecture comprises three tightly coupled components designed to achieve synergistic optimization. The first component transforms images from heterogeneous modalities into a unified feature space through learned mapping functions f_r and f_p, yielding modality-invariant representations $\mathbf{Z}_x = f_r(I_x)$ and $\mathbf{Z}_y = f_p(I_y)$, where $\mathbf{Z}_x, \mathbf{Z}_y \in \mathbb{R}^{C\times H\times W}$. This transformation is guided by a similarity-preserving objective that enforces consistent representations for corresponding spatial locations while maintaining discriminability for non-corresponding regions.

The second component performs geometric alignment by partitioning the shared feature maps into spatial blocks and establishing dense correspondences through displacement field regression. For each block centered at position (i, j), heterogeneous node features $\mathbf{h}_r^{(i,j)}$ and $\mathbf{h}_p^{(i,j)}$ are constructed by fusing local appearance descriptors with spatial position encodings. These nodes form a graph structure where message passing aggregates contextual information from neighboring correspondences, enabling the prediction of displacement vectors $(\Delta\hat{x}_{(i,j)}, \Delta\hat{y}_{(i,j)})$ that capture both local deformations and global geometric constraints. The predicted displacement field is then applied through differentiable coordinate transformation to obtain aligned image pairs $\hat{I}_x^R$ and $\hat{I}_y^R$.

The third component extracts change information from the aligned images through multiscale feature aggregation and cross-modal attention fusion. Hierarchical features $\mathbf{F}_x^{agg}, \mathbf{F}_y^{agg} \in \mathbb{R}^{C\times H\times W}$ are extracted and associated through a key-value attention mechanism that computes bidirectional attention weights $\mathbf{A}_{x\to y}$ and $\mathbf{A}_{y\to x}$. The attention-weighted features are fused to form the change-enhanced representation $\mathbf{F}_{change}$, which is decoded to produce the change probability map $\mathbf{P} \in \mathbb{R}^{1\times H\times W}$.

The key innovation of the framework lies in its joint optimization strategy that couples registration and change detection through shared feature representations and complementary loss functions. The registration module receives gradient signals from the change detection loss, enabling it to prioritize alignment quality in regions critical for accurate change identification. Conversely, the change detection module benefits from geometrically consistent input features, reducing false alarms caused by misregistration artifacts. During supervised training, the available binary change annotation $\mathbf{Y}$ also defines an unchanged-region reliability map $\mathbf{M} = \mathbf{1} - \mathbf{Y}$ (resampled consistently at each feature scale). $\mathbf{M}$ weights only the appearance-based contrastive and normalized cross correlation (NCC) terms. It does not gate the displacement target or graph regularizer. Therefore, changed pixels cannot reduce the loss by being warped onto unrelated content, while the predicted displacement field remains spatially coherent. This is a training-time use of the same annotation already required by $\mathcal{L}_{change}$, not an additional a priori mask; no change mask or loss evaluation is required by the registration branch at inference.

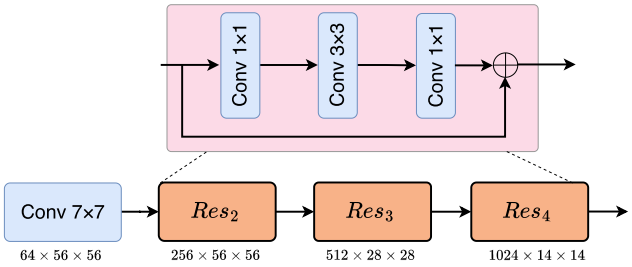


Fig. 3. Architecture of the backbone feature extraction network with residual blocks and bottleneck structures.

B. Similarity Contrast Spatial Mapping

To address the feature heterogeneity inherent in cross-modal imagery, we design a modality-bridging mechanism that projects images from different sensors into a shared representation space where geometric correspondence can be reliably established. The backbone network, illustrated in Fig. 3, implements the mapping functions f_r and f_p through a hierarchical architecture consisting of an initial 7×7 convolutional layer followed by three residual stages (Res_2–Res_4).

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Each residual stage employs bottleneck structures with skip connections, progressively transforming feature dimensions from $64 \times 56 \times 56$ to $1024 \times 14 \times 14$.

The core principle underlying our feature mapping is that corresponding spatial locations across modalities should exhibit similar representations in the learned space, while noncorresponding locations should remain distinguishable. Let $\{(r_i, p_i)\}_{i=1}^{N}$ now denote sampled co-located feature patches and let $m_i \in [0, 1]$ be the average of the unchanged-region map $\mathbf{M}$ over patch i. The appearance contrast is evaluated only where a valid unchanged correspondence exists

$$\mathcal{L}_{contrast}^{M} = \frac{\sum_{i=1}^{N} m_i \left[\mathcal{L}_r(i) + \mathcal{L}_p(i)\right]}{\sum_{i=1}^{N} m_i + \varepsilon} \tag{1}$$

where

$$\mathcal{L}_r(i) = -\log \frac{\exp\left(s\left(f_r(r_i), f_p(p_i)\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(s\left(f_r(r_i), f_p\left(p_j\right)\right)/\tau\right)} \tag{2}$$

$$\mathcal{L}_p(i) = -\log \frac{\exp\left(s\left(f_p(p_i), f_r(r_i)\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(s\left(f_p(p_i), f_r\left(r_j\right)\right)/\tau\right)} \tag{3}$$

where $s(\cdot,\cdot)$ computes cosine similarity and τ controls the concentration of the distribution. The reliability weight is not used to erase changed locations from the feature maps; it only prevents those locations from acting as positive appearance correspondences. Consequently, change evidence remains available to the detection decoder.

To enhance spatial discriminability, we incorporate position-aware feature refinement that generates attention weights $\mathbf{A} \in \mathbb{R}^{1\times H\times W}$ for each feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$

$$\mathbf{A} = \sigma\left(\mathbf{W}_2 \mathrm{ReLU}\left(\mathbf{W}_1 \mathbf{F}\right)\right), \quad \mathbf{F}' = \mathbf{A} \odot \mathbf{F} \tag{4}$$

where $\mathbf{W}_1 \in \mathbb{R}^{C/r\times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{1\times C/r}$ are learnable parameters. This mechanism enables the network to emphasize spatially informative regions while suppressing background clutter, providing robust features for subsequent correspondence estimation.

C. Matchpoint Displacement Regression Module

Accurate geometric alignment requires establishing dense correspondences that capture both local deformations and global transformation constraints. We formulate this as a structured prediction problem where displacement vectors are regressed for a regular grid of matchpoints, with spatial relationships among neighboring points explicitly modeled to ensure geometric consistency.

The feature maps $\mathbf{F}_r, \mathbf{F}_p \in \mathbb{R}^{C\times H\times W}$ are partitioned into 16×16 blocks, with block centers serving as matchpoints at coordinates

$$x_{(i,j)} = (j - 0.5) \times 16, \quad y_{(i,j)} = (i - 0.5) \times 16. \tag{5}$$

For each matchpoint, we construct a hybrid descriptor that integrates local appearance with global position awareness. Let $\mathbf{f}_r^{(i,j)}, \mathbf{f}_p^{(i,j)} \in \mathbb{R}^{C}$ denote the aggregated features within each block. Position encodings $\mathbf{e}_{(i,j)} \in \mathbb{R}^{C'}$ are computed through

$$\mathbf{e}_{(i,j)} = \phi\left(\mathbf{W}_{pos}\left[x_{(i,j)}, y_{(i,j)}\right]^{T}\right). \tag{6}$$

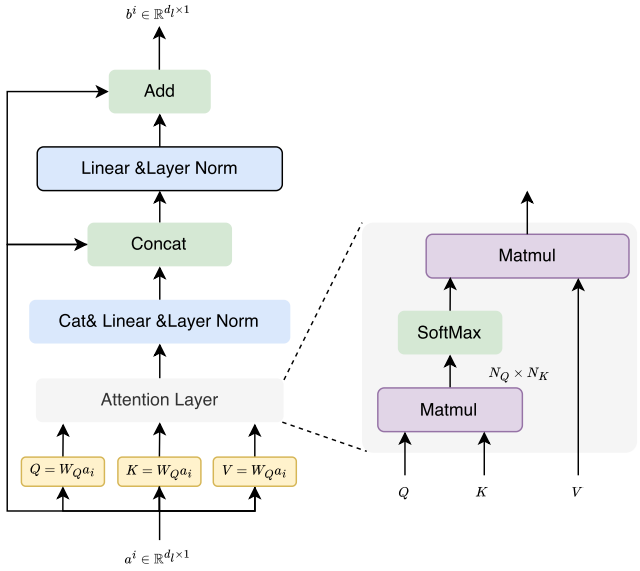


Fig. 4. Cross-modal attention mechanism architecture with Q–K–V transformations and residual connections.

The hybrid descriptors are formed as $\mathbf{h}_r^{(i,j)} = [\mathbf{f}_r^{(i,j)}, \mathbf{e}_{(i,j)}]$ and $\mathbf{h}_p^{(i,j)} = [\mathbf{f}_p^{(i,j)}, \mathbf{e}_{(i,j)}]$.

These matchpoints are organized as nodes in a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where edges connect spatially adjacent points. The graph structure enables propagation of geometric constraints through iterative message passing

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{7}$$

where $\tilde{\mathbf{A}}$ is the adjacency matrix with self-loops, $\tilde{\mathbf{D}}$ is the corresponding degree matrix, and $\mathbf{W}^{(l)}$ contains learnable parameters. This formulation allows each matchpoint to aggregate information from its spatial neighborhood, enforcing local smoothness while accommodating nonrigid deformations.

To enhance cross-modal correspondence modeling, we incorporate a cross-modal attention mechanism within the graph processing pipeline, as shown in Fig. 4. For each node feature, query–key–value projections enable selective aggregation of information from corresponding locations in the other modality. The attention-weighted features undergo residual combination with the original representations, yielding refined descriptors that capture both intramodal structure and intermodal correspondence.

The displacement prediction for each matchpoint is obtained through

$$(\Delta x_{(i,j)}, \Delta y_{(i,j)}) = g_\theta\Big(\mathbf{h}_r^{(i,j)}, \mathbf{h}_p^{(i,j)} \times \{\mathbf{h}_r^{(m,n)}, \mathbf{h}_p^{(m,n)}\}_{(m,n)\in\mathcal{N}_{(i,j)}}\Big). \tag{8}$$

The supervised displacement term minimizes the discrepancy between predicted and ground-truth geometric displacements

$$\mathcal{L}_{disp} = \sum_{i,j} \left\|\Delta\left(x_{(i,j)}, y_{(i,j)}\right) - \left(\Delta\hat{x}_{(i,j)}, \Delta\hat{y}_{(i,j)}\right)\right\|_2^2 \tag{9}$$

where the target Δ is generated by the recorded synthetic/projective transform (see Section IV-A) and therefore

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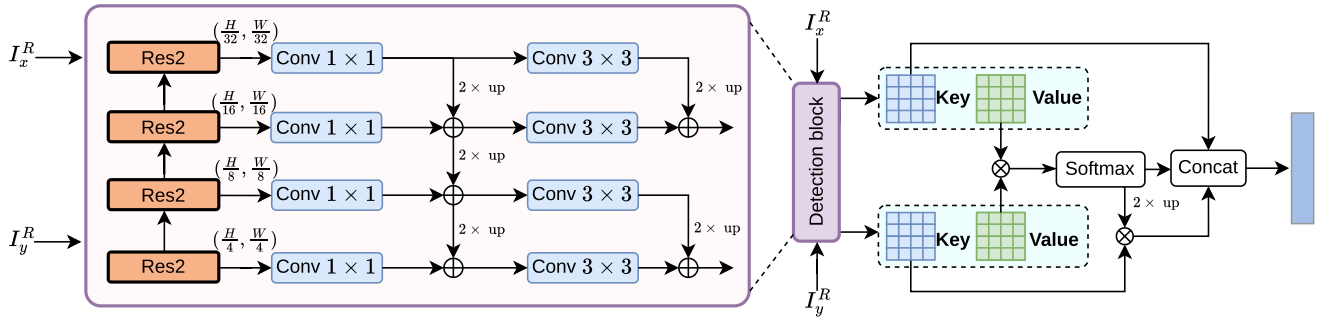


Fig. 5. Change detection module architecture with multiscale feature extraction and cross-modal attention fusion.

remains valid at every grid point, including a point whose scene content changed. It supervises coordinates, not intertemporal radiometry. Graph regularity is likewise imposed on the displacement vectors rather than on image intensities. With $\mathbf{d}_u$ denoting the predicted displacement at graph node u, we write

$$\mathcal{L}_{smooth} = \frac{1}{|\mathcal{E}|} \sum_{(u,v)\in\mathcal{E}} \|\mathbf{d}_u - \mathbf{d}_v\|_1 . \tag{10}$$

This term encodes the assumption that the sensor/viewpoint-induced deformation varies locally smoothly; it does not encourage a changed building to imitate its previous appearance.

The only image-level registration term is a change-excluded NCC. Let $I_y^R = \mathcal{W}(I_y, \mathbf{d})$, let M_{ij} be the unchanged reliability at pixel (i, j), and define $\mu_x^M = \sum M_{ij} I_x(i, j) / \left(\sum M_{ij} + \varepsilon\right)$ and μ_y^M analogously. For compactness, set $\widetilde{I}_x = I_x - \mu_x^M$ and $\widetilde{I}_y^R = I_y^R - \mu_y^M$. We use

$$\begin{aligned} \mathcal{L}_{\mathrm{NCC}}^M &= -\frac{N_M}{D_M + \varepsilon} \\ N_M &= \sum_{i,j} M_{ij} \widetilde{I}_x(i, j)\, \widetilde{I}_y^R(i, j) \\ D_M &= \sqrt{\sum_{i,j} M_{ij} \widetilde{I}_x(i, j)^2} \sqrt{\sum_{i,j} M_{ij} \widetilde{I}_y^R(i, j)^2}. \end{aligned} \tag{11}$$

The registration objective is thus

$$\mathcal{L}_{reg} = \mathcal{L}_{disp} + \eta \mathcal{L}_{smooth} + \epsilon \mathcal{L}_{\mathrm{NCC}}^M. \tag{12}$$

This decomposition makes the support of each constraint explicit: appearance fidelity is restricted to unchanged pixels, whereas displacement supervision and field regularity remain dense. The predicted displacement field is interpolated to full resolution and applied through differentiable coordinate transformation to produce the aligned images, enabling end-to-end gradient flow from the change detection objective back to the registration parameters.

D. Change Detection Module

Following geometric alignment, accurate change detection requires effective integration of multiscale semantic information and discriminative comparison of the aligned feature representations. The change detection module, illustrated in Fig. 5, employs hierarchical feature extraction followed by cross-modal attention-based fusion to identify changed regions.

The aligned images $\hat{I}_x^R$ and $\hat{I}_y^R$ are processed through a multiresolution feature pyramid that extracts representations at scales $(H/32, W/32)$, $(H/16, W/16)$, $(H/8, W/8)$, and $(H/4, W/4)$

$$\mathbf{F}_x^{(l)}, \mathbf{F}_y^{(l)} \in \mathbb{R}^{C_l \times \frac{H}{s_l} \times \frac{W}{s_l}}, \quad l = 1, \ldots, L. \tag{13}$$

These multiscale features are progressively fused through channel compression via 1×1 convolutions, $2\times$ upsampling, and 3×3 refinement convolutions, yielding aggregated representations

$$\mathbf{F}_x^{agg} = \Psi\left(\{\mathbf{F}_x^{(l)}\}_{l=1}^L\right), \quad \mathbf{F}_y^{agg} = \Psi\left(\{\mathbf{F}_y^{(l)}\}_{l=1}^L\right). \tag{14}$$

To capture discriminative change patterns, we employ bidirectional cross-modal attention that computes correspondence-aware feature transformations. Key and value projections are generated through

$$\mathbf{K}_x = f_K\left(\mathbf{F}_x^{agg}\right), \quad \mathbf{V}_x = f_V\left(\mathbf{F}_x^{agg}\right) \tag{15}$$

$$\mathbf{K}_y = f_K\left(\mathbf{F}_y^{agg}\right), \quad \mathbf{V}_y = f_V\left(\mathbf{F}_y^{agg}\right). \tag{16}$$

Attention weights are computed through normalized inner products

$$\mathbf{A}_{x\to y}(i, j) = \frac{\exp\left(\mathbf{K}_x(i) \cdot \mathbf{K}_y(j)\right)}{\sum_{j'} \exp\left(\mathbf{K}_x(i) \cdot \mathbf{K}_y(j')\right)}. \tag{17}$$

The attention-weighted feature aggregation produces

$$\mathbf{F}'_x(i) = \sum_j \mathbf{A}_{x\to y}(i, j)\, \mathbf{V}_y(j) \tag{18}$$

$$\mathbf{F}'_y(i) = \sum_j \mathbf{A}_{y\to x}(i, j)\, \mathbf{V}_x(j). \tag{19}$$

The bidirectional attention mechanism serves a dual purpose: for unchanged regions, the attention focuses on corresponding locations, producing similar transformed features; for changed regions, the attention distribution becomes diffuse as no strong correspondence exists, amplifying the feature discrepancy. This behavior emerges naturally from the attention formulation without explicit supervision, providing an intrinsic mechanism for change enhancement.

The transformed features are concatenated to form the change representation

$$\mathbf{F}_{change} = \left[\mathbf{F}'_x, \mathbf{F}'_y\right] \in \mathbb{R}^{2C \times H \times W}. \tag{20}$$

The final change probability map is produced through

$$\mathbf{P} = \sigma\left(g_{dec}\left(\mathbf{F}_{change}\right)\right) \tag{21}$$

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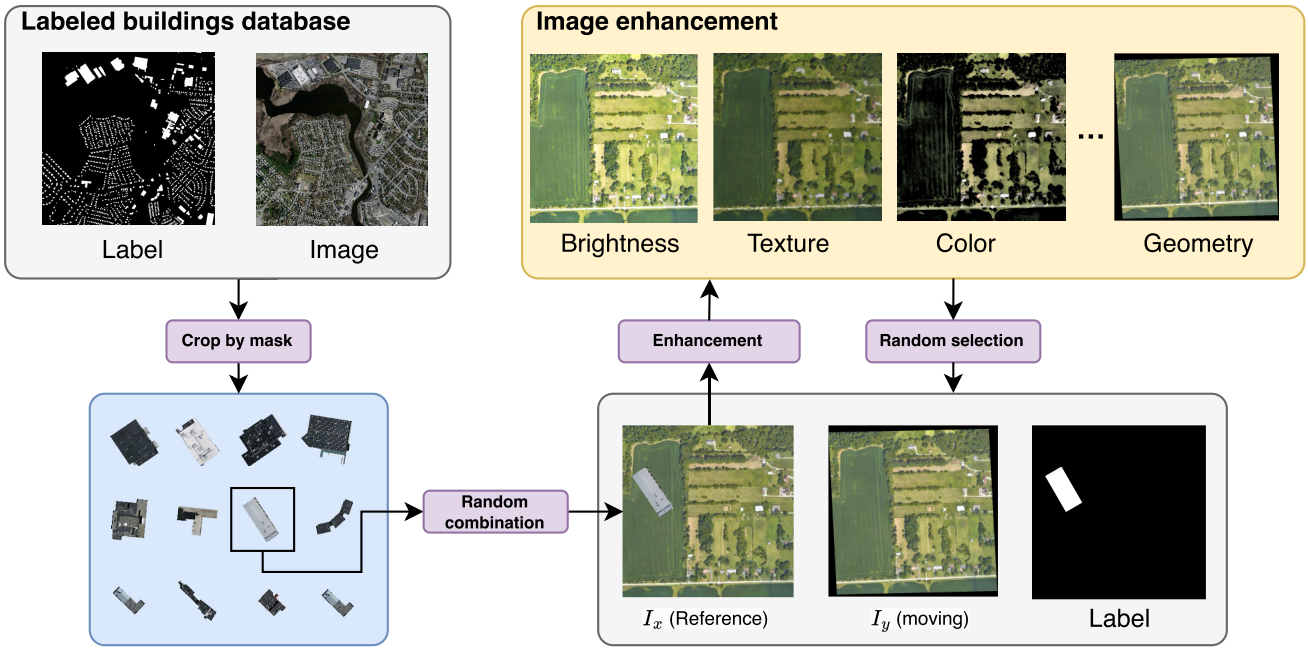


Fig. 6. Process of generating a training dataset for remote sensing image change detection without image registration.

where $g_{dec}(\cdot)$ denotes the decoding network and $\sigma(\cdot)$ is the sigmoid activation. The detection loss employs binary cross-entropy between $\mathbf{P}$ and ground-truth labels, with gradients propagating through the entire pipeline to jointly optimize all components.

The binary cross-entropy loss for change detection is formulated as follows:

$$\mathcal{L}_{\text{change}} = -\frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\Big[Y_{i,j}\log P_{i,j} + \left(1-Y_{i,j}\right)\log\left(1-P_{i,j}\right)\Big] \quad (22)$$

where $Y_{i,j} \in \{0,1\}$ denotes the ground-truth label at pixel (i,j) and $P_{i,j}$ is the predicted change probability. This loss is minimized during training to supervise the change detection output, with gradients flowing back through the entire network to jointly optimize all modules.

IV. Experiment and Results

A. Implementation Details

To generate sufficient training samples for learning registration-aware change detection (RACD) without pre-aligned image pairs, we develop a data synthesis pipeline based on building detection datasets, as illustrated in Fig. 6. The synthetic pipeline, though primarily based on building samples, is designed to instill in the model a generalizable ability to distinguish genuine semantic changes from various geometric misalignments. The diversity of augmentations ensures effective transfer to other change types. The pipeline begins with a labeled building database containing binary mask images and corresponding high-resolution remote sensing imagery. Image patches containing individual buildings are extracted using label masks, then subjected to augmentation operations including brightness adjustment, texture enhancement, color alteration, and geometric transformations. The augmented patches are randomly combined to generate synthetic image pairs, where I_x serves as the reference image and I_y contains spatially displaced or altered buildings. Corresponding ground-truth labels are simultaneously generated to mark changed building positions. This synthesis strategy substantially increases sample diversity, enhancing the model's robustness and generalization capability for handling real-world unregistered imagery.

We evaluate the proposed method, termed RACD, on four datasets spanning different sensors, resolutions, and application scenarios. The WHU-CD dataset comprises two aerial images of Christchurch at 0.075-m resolution, acquired in 2012 and 2016 to assess post-earthquake reconstruction. The original 32 507 × 15 354 pixel images are cropped into nonoverlapping 224 × 224 patches and divided into training, validation, and test sets at an 8:1:1 ratio. The CDD dataset contains 11 bitemporal image pairs with spatial resolutions ranging from 3 to 100 cm/pixel, targeting changes in buildings and roads while excluding seasonal variations. LEVIR-CD+ is a large-scale building change detection dataset comprising 1970 samples of 1024 × 1024 pixels at 0.5-m resolution, with 985 samples for training and 985 for testing, encompassing approximately 80 000 changed building instances. The Greenhouse Change Detection dataset (GHCD) consists of 300 time-series Google Earth images of 1024 × 1024 pixels at 0.5-m resolution with manually annotated greenhouse change labels.

To systematically evaluate the joint registration and change detection capability under varying degrees of geometric misalignment, we design experiments addressing both simulated and real-world unregistered scenarios. For the first three benchmark datasets (WHU-CD, CDD, and LEVIR-CD+), where original image pairs are precisely

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TABLE I
TRAINING CONFIGURATION FOR RACD MODULES ACROSS FOUR DATASETS

Dataset	Module	Patch Size	Batch Size	Epochs	Learning Rate
WHU-CD	Similarity Mapping	224×224	32	100	1e-3 → 1e-5 (cosine)
	Displacement Regression	512×512	16	80	1e-4 → 1e-6 (step)
	Change Detection	224×224	24	120	1e-3 → 1e-5 (polynomial)
CDD	Similarity Mapping	256×256	28	150	1e-3 → 1e-5 (cosine)
	Displacement Regression	512×512	12	100	1e-4 → 1e-6 (step)
	Change Detection	256×256	20	180	1e-3 → 1e-5 (polynomial)
LEVIR-CD+	Similarity Mapping	256×256	24	80	1e-3 → 1e-5 (cosine)
	Displacement Regression	1024×1024	8	60	1e-4 → 1e-6 (step)
	Change Detection	256×256	16	100	1e-3 → 1e-5 (polynomial)
GHCD	Similarity Mapping	512×512	16	120	1e-3 → 1e-5 (cosine)
	Displacement Regression	1024×1024	6	80	1e-4 → 1e-6 (step)
	Change Detection	512×512	12	150	1e-3 → 1e-5 (polynomial)

registered, we apply controlled projective transformations to one image in each pair to simulate geometric misalignment. The transformation parameters are randomly sampled within predefined ranges and recorded as ground-truth alignment references for quantitative evaluation of registration accuracy. This controlled simulation enables systematic analysis of method performance across different misalignment magnitudes. For the GHCD dataset, which contains naturally unregistered image pairs without strict geometric alignment, we directly input the original imagery to evaluate performance under authentic application conditions. This dual evaluation strategy—combining controlled simulation with real-world unregistered data—provides comprehensive assessment of the RACD method's effectiveness across diverse operational scenarios.

Table I summarizes the training configuration for each RACD module across the four datasets, including patch dimensions, batch sizes, training epochs, and learning rate schedules. Algorithm 1 provides an overview of the complete training and inference procedure. The unchanged reliability map is obtained directly from the training change label that is already consumed by the detection loss; Fig. 6 shows that the synthetic pipeline generates this label together with each image pair. Hence, it introduces neither a separately annotated registration mask nor an inference-time prerequisite. At test time, all loss-only quantities (Y, M, $\mathcal{L}^{M}_{contrast}$, and $\mathcal{L}^{M}_{\mathrm{NCC}}$) are removed, and RACD predicts the displacement and change map in a single forward pass. The displacement regression module employs three graph convolutional layers with 64 hidden units per layer, and three message-passing iterations are performed during both training and inference. This configuration is fixed across all datasets and was chosen to balance registration accuracy with computational efficiency. All modules are optimized using Adam with a progressive training strategy that pretrains on smaller patches before fine-tuning on larger scales to improve generalization across varying spatial extents. Hyperparameters are determined through cross-validation, yielding final settings of $\tau = 1$, $\delta = 10$, $\eta = 0.001$, and $\epsilon = 0.1$. Experiments are conducted on hardware comprising an NVIDIA GeForce GTX Titan X GPU and Intel Core i5-1135G7 CPU.

Algorithm 1 RACD Training and Inference Procedure

Require: Image pairs $\{(I^i_x, I^i_y)\}_{i=1}^N$, change labels $\{Y^i\}_{i=1}^N$, recorded geometric transforms $\{\Delta^i\}_{i=1}^N$
Ensure: Trained model parameters θ
Training:
Initialize $\theta_r, \theta_p, \theta_{gcn}, \theta_{cd}$; Set $\tau = 1$, $\delta = 10$, $\eta = 0.001$, $\epsilon = 0.1$
for epoch $e = 1$ to E_{max} **do**
  for each batch $\{(I^b_x, I^b_y, Y^b)\}$ **do**
    $\mathbf{M}^b = \mathbf{1} - \mathbf{Y}^b$; resample $\mathbf{M}^b$ at the active feature scales
    $\mathbf{Z}_x = f_r(I^b_x)$, $\mathbf{Z}_y = f_p(I^b_y)$ {Feature mapping}
    Compute $\mathcal{L}^{M}_{contrast}$ via Eqs. (1)–(3)
    Partition features into blocks; construct graph $\mathcal{G}$
    Predict $(\Delta\hat{x}_{(i,j)}, \Delta\hat{y}_{(i,j)})$; compute $\mathcal{L}_{disp}$ and $\mathcal{L}_{smooth}$
    $\hat{I}^R_x, \hat{I}^R_y = \mathrm{STN}(I^b_x, I^b_y, \{\Delta\hat{x}, \Delta\hat{y}\})$
    Compute change-excluded $\mathcal{L}^{M}_{\mathrm{NCC}}$ and $\mathcal{L}_{reg}$ via Eq. (12)
    Extract $\mathbf{F}^{agg}_x, \mathbf{F}^{agg}_y$; compute attention $\mathbf{A}_{x\leftrightarrow y}$
    $\mathbf{P} = \sigma(g_{dec}([\mathbf{F}'_x, \mathbf{F}'_y]))$; compute $\mathcal{L}_{change}$
    $\mathcal{L}_{total} = \mathcal{L}^{M}_{contrast} + \lambda_1\mathcal{L}_{reg} + \lambda_2\mathcal{L}_{change}$
    Update $\theta \leftarrow \theta - \alpha\nabla_\theta\mathcal{L}_{total}$
  end for
end for
Inference:
Map features $\mathbf{Z}^{test}_x, \mathbf{Z}^{test}_y$; predict displacements without Y or M
Register: $\hat{I}^R_x, \hat{I}^R_y = \mathrm{STN}(I^{test}_x, I^{test}_y, \{\Delta\hat{x}, \Delta\hat{y}\})$
return: $\mathbf{P}^{test} = \sigma(g_{dec}(\mathbf{F}^{test}_{change}))$

B. Image Registration Evaluation

Fig. 7 presents qualitative registration results across the four datasets. Each visualization comprises two subfigures: the left displays ground-truth alignment through transparency overlay of reference and target images, while the right shows the alignment achieved by each method, where deviations between blue and red boundary lines indicate registration errors. SIFT [54], employing scale-invariant local feature descriptors, exhibits the largest alignment errors due to its limited capacity for handling complex geometric distortions in

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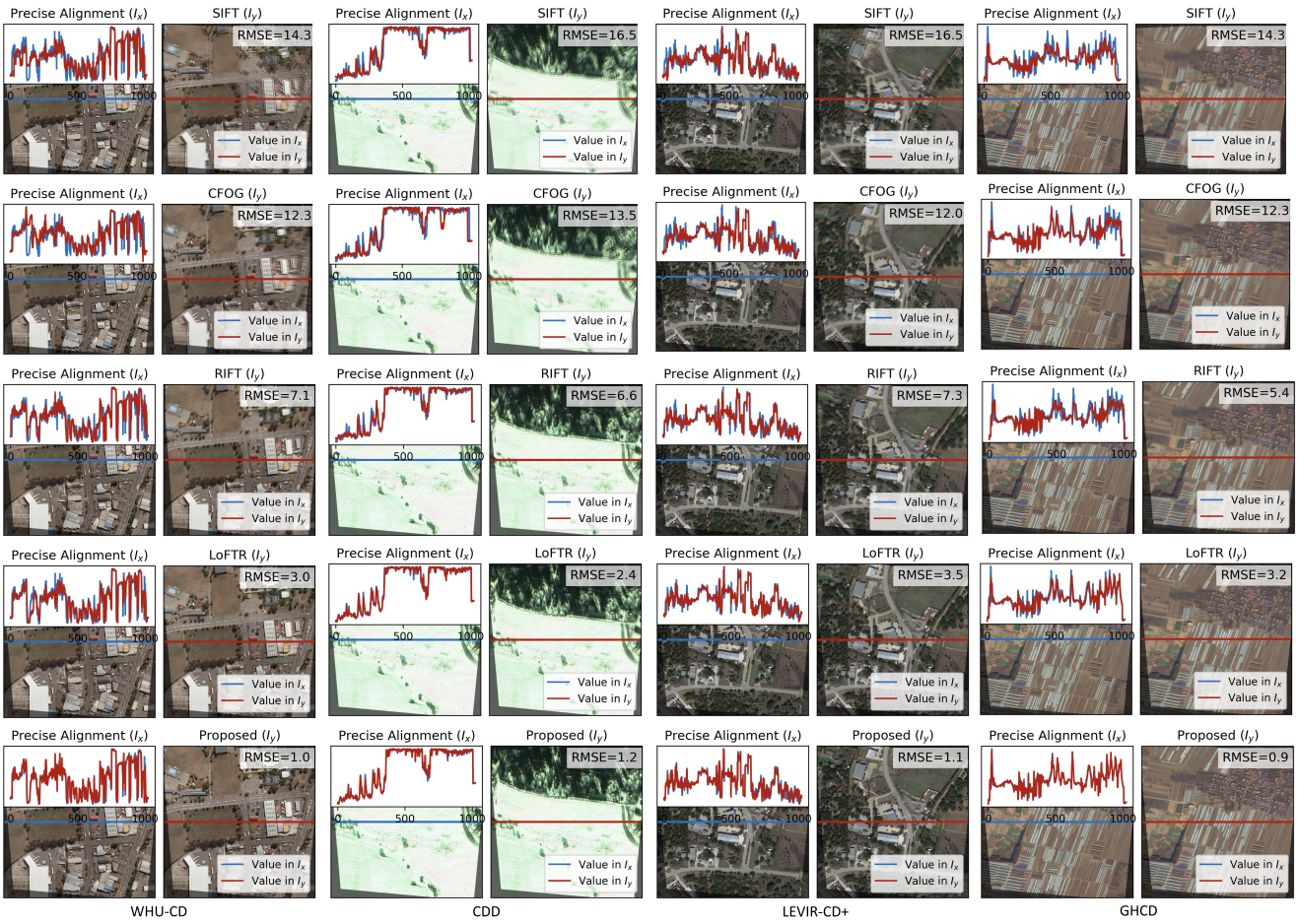


Fig. 7. Qualitative matching results of different methods on four datasets. Left subfigures show ground-truth alignment with transparency overlay; right subfigures display alignment after each method's registration, where deviations between blue and red lines indicate matching errors.

remote sensing imagery. CFOG [55], which constructs feature descriptors based on gradient orientation histograms within local neighborhoods, achieves improved matching through enhanced robustness to illumination variations, though notable deviations persist under significant geometric transformations. RIFT [56], utilizing phase congruency-based feature detection combined with maximum index maps for description, demonstrates further improvements by exploiting radiation-invariant structural information. LoFTR [57], which employs attention mechanisms for dense feature matching without explicit keypoint detection, achieves the second-best performance by establishing reliable correspondences through global context aggregation. The proposed method consistently produces the most accurate registration across all datasets, validating the effectiveness of combining graph-based correspondence reasoning with joint optimization for change detection.

To quantitatively evaluate registration accuracy, we define four corner points in the reference image as evaluation benchmarks: top-left $(100, 100)$, bottom-left $(100, 924)$, bottom-right $(924, 924)$, and top-right $(924, 100)$, denoted as $\{(u_s^k, v_s^k)\}_{k=1}^4$. Let $\{(u_t^k, v_t^k)\}_{k=1}^4$ represent corresponding target points, and $\{(\hat{u}_s^k, \hat{v}_s^k)\}_{k=1}^4$ denote transformed benchmark points. It is worth noting that the four corner points are deliberately chosen as the evaluation benchmarks, as they represent the most geometrically sensitive locations in the image. Under common parametric transformation models, alignment accuracy at these extreme positions serves as a reliable indicator of global registration quality. All competing methods are evaluated under the identical protocol to ensure fair comparison. Moreover, the selected coordinates are defined relative to the image size and are consistently applied across all datasets, providing a standardized and reproducible evaluation framework. Registration accuracy is measured using root-mean-square error (RMSE), mean absolute error (MAE), and mean Euclidean error (MEE).

Table II reports quantitative results across all datasets, with best performance highlighted in bold. The proposed method achieves superior performance across all datasets and metrics, demonstrating consistent advantages in geometric alignment accuracy. On WHU-CD, our method attains RMSE, MAE, and MEE of 1.73, 3.45, and 1.37 pixels, respectively, substantially outperforming all comparison methods. This improvement stems from the joint optimization strategy that enables registration parameters to be refined based on change detection feedback, producing alignment specifically optimized for downstream semantic analysis rather than generic similarity metrics. The handcrafted feature-based methods (SIFT, CFOG, and RIFT) exhibit progressively improved performance, yet their reliance on predefined descriptors limits adaptability to the diverse appearance variations in remote sensing imagery. LoFTR achieves competitive results through learned dense

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TABLE II
QUANTITATIVE REGISTRATION EVALUATION ON FOUR DATASETS (PIXELS)

Method	RMSE ↓	MAE ↓	MEE ↓
WHU-CD			
SIFT	23.67 ± 5.24	48.92 ± 11.37	20.14 ± 4.69
CFOG	8.43 ± 3.18	17.29 ± 6.71	7.56 ± 2.94
RIFT	5.96 ± 2.47	12.35 ± 5.28	5.28 ± 2.31
LoFTR	3.52 ± 1.86	7.14 ± 3.92	3.08 ± 1.75
Proposed	$\mathbf{1.73 \pm 0.92}$	$\mathbf{3.45 \pm 1.88}$	$\mathbf{1.37 \pm 0.79}$
CDD			
SIFT	18.39 ± 4.51	37.84 ± 9.47	16.72 ± 4.25
CFOG	7.26 ± 2.83	14.91 ± 5.92	6.58 ± 2.64
RIFT	5.14 ± 2.19	10.63 ± 4.62	4.61 ± 2.06
LoFTR	2.95 ± 1.57	5.98 ± 3.29	2.60 ± 1.49
Proposed	$\mathbf{1.42 \pm 0.76}$	$\mathbf{2.82 \pm 1.56}$	$\mathbf{1.16 \pm 0.67}$
LEVIR-CD+			
SIFT	9.84 ± 3.27	20.25 ± 6.86	8.93 ± 3.09
CFOG	4.61 ± 2.09	9.47 ± 4.36	4.18 ± 1.96
RIFT	3.38 ± 1.64	6.98 ± 3.47	3.02 ± 1.54
LoFTR	2.07 ± 1.18	4.19 ± 2.45	1.82 ± 1.10
Proposed	$\mathbf{0.98 \pm 0.57}$	$\mathbf{1.94 \pm 1.15}$	$\mathbf{0.81 \pm 0.50}$
GHCD			
SIFT	15.61 ± 4.08	32.13 ± 8.55	14.19 ± 3.84
CFOG	6.37 ± 2.55	13.09 ± 5.33	5.77 ± 2.39
RIFT	4.52 ± 1.97	9.34 ± 4.17	4.05 ± 1.85
LoFTR	2.64 ± 1.39	5.35 ± 2.89	2.32 ± 1.30
Proposed	$\mathbf{1.25 \pm 0.68}$	$\mathbf{2.48 \pm 1.38}$	$\mathbf{1.03 \pm 0.60}$

matching, though its task-agnostic training objective prevents exploitation of change detection supervision for registration refinement.

To validate the contribution of individual loss components, we conduct ablation experiments focusing on the local mean squared error (mse) loss and global NCC loss. The mse loss measures the Euclidean distance between transformed and target corner points

$$\mathcal{L}_{\text{MSE}} = \frac{1}{4}\sum_{k=1}^{4}\left[\left(\hat{u}_s^k - u_t^k\right)^2 + \left(\hat{v}_s^k - v_t^k\right)^2\right]. \tag{23}$$

This loss enables rapid convergence by directly supervising local feature alignment. However, as illustrated in Fig. 8, optimizing mse alone produces imbalanced corner accuracy, with certain points exhibiting substantially larger errors due to the network prioritizing easily optimized correspondences.

The NCC component provides global registration supervision only over valid unchanged support. Specifically, the ablation uses the change-excluded $\mathcal{L}_{\text{NCC}}^{M}$ in (11), rather than an unweighted correlation over the entire image. Its masked means and covariance are computed from $M_{ij} > 0$, so changed buildings, demolitions, and other noncorresponding content contribute zero appearance-alignment gradient. Within the unchanged support, NCC distributes optimization effort more uniformly across spatial locations and complements the local coordinate mse.

As demonstrated in Fig. 8, combining both losses achieves lower registration errors in later training stages compared to mse alone. Fig. 9 illustrates the improved corner alignment

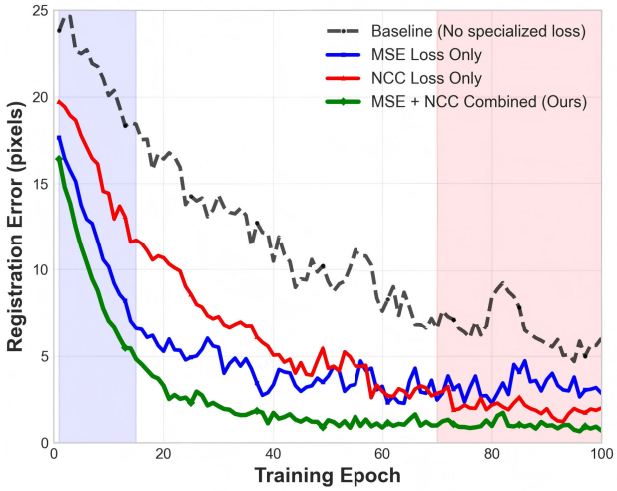


Fig. 8. Ablation study on registration loss functions: training dynamics and corner point alignment accuracy.

consistency when incorporating NCC loss, with previously problematic corners exhibiting substantially reduced errors. The complementary nature of these losses—local coordinate mse accelerating convergence and change-excluded NCC balancing unchanged-region alignment—establishes robust geometry without rewarding false alignment inside truly changed regions.

C. Change Detection Evaluation

To comprehensively evaluate the change detection performance of the proposed method, we compare against state-of-the-art approaches including: IFN [58], BiT [59], TransUNetCD [60], DARNet [61], and SwinSUNet [44]. All comparison methods are evaluated under identical experimental settings using their official implementations with recommended hyperparameters.

Fig. 10 presents a qualitative comparison across representative examples from the four datasets. White regions indicate correctly identified changes, red regions denote false positives, and green regions represent missed detections. The proposed method consistently produces change maps with more precise boundaries and fewer artifacts compared to competing approaches. On WHU-CD, our method accurately captures subtle building contour changes while suppressing false alarms in complex backgrounds. For CDD, the method effectively distinguishes genuine object changes from pseudo-changes induced by seasonal variations. On LEVIR-CD+, superior boundary preservation is evident in dense urban areas with complex building geometries. For GHCD, the method demonstrates robust detection of greenhouse structural changes including construction, demolition, and expansion.

Quantitative evaluation employs precision (P), recall (R), intersection over union (IoU), and $F1$-score as metrics. Results presented in Fig. 11 demonstrate that the proposed method achieves optimal or near-optimal performance across all datasets. On WHU-CD, the proposed method attains 95.48% precision, 95.26% recall, 85.56% IoU, and a 95.14% $F1$-score. For CDD, it achieves 92.32% precision, 91.92% recall, 82.11% IoU, and a 92.09% $F1$-score, demonstrating robustness to

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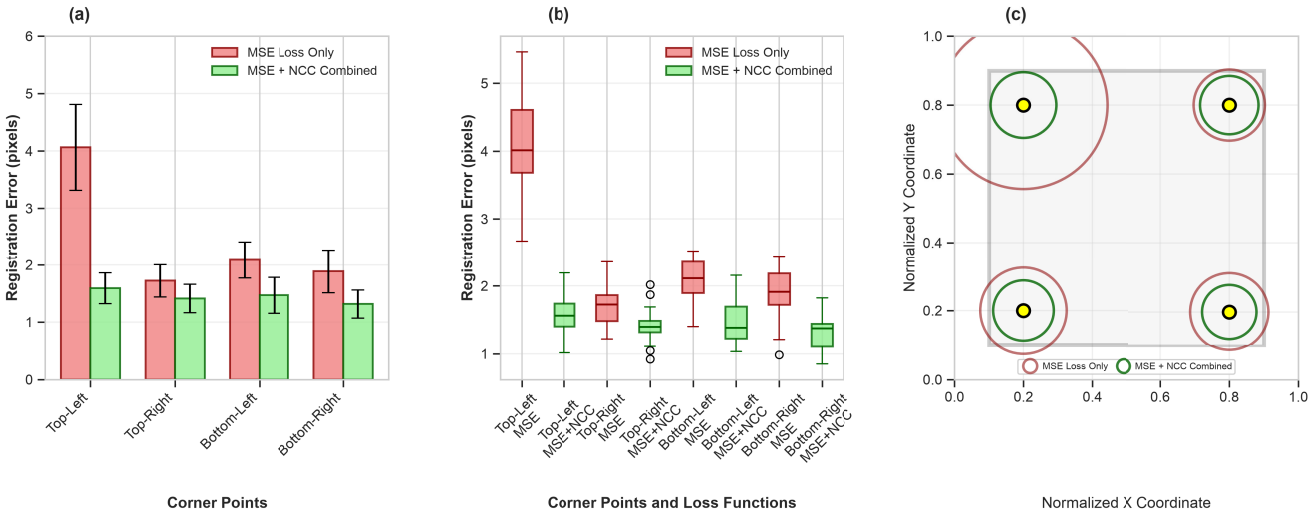


Fig. 9. Corner point alignment after incorporating combined mse and NCC losses. (a) Corner point registration error. (b) Error distribution analysis. (c) Spatial error visualization.

TABLE III
ABLATION STUDY ON LEVIR-CD+ DATASET

Model Configuration	P (%)	R (%)	IoU (%)	F1-Score (%)
Joint Registration vs. Direct Detection				
Direct CD (pre-registered input)	94.12	93.28	88.15	93.70
Direct CD (unregistered input)	87.43	85.67	76.24	86.54
Sequential (separate reg. + CD)	93.85	93.02	87.68	93.43
Joint optimization (proposed)	**95.83**	**95.10**	**91.35**	**95.46**
Component Contribution Analysis				
Baseline encoder-decoder	93.24	92.51	86.79	92.87
+ Adversarial training	94.37	93.65	88.92	94.01
+ Long-range attention	95.19	94.46	90.21	94.82
+ Multi-scale fusion	95.56	94.83	90.78	95.19
+ Cross-modal attention	95.83	95.10	91.35	95.46

seasonal variation interference. On LEVIR-CD+, precision, recall, IoU, and $F1$-score reach 98.03%, 98.02%, 91.59%, and 97.99%, respectively, indicating strong performance in large-scale building change detection. For GHCD, the proposed method obtains 92.09% precision, 91.25% recall, 79.56% IoU, and a 91.78% $F1$-score, demonstrating good generalization to domain-specific applications.

The proposed method achieves balanced precision and recall across all datasets, indicating accurate change identification with minimal missed detections. Performance advantages are particularly pronounced on challenging datasets (CDD and GHCD), demonstrating enhanced robustness in complex scenarios where geometric misalignment and appearance variations pose significant challenges for conventional methods.

To systematically validate the contribution of joint optimization and individual components, we conduct comprehensive ablation experiments on LEVIR-CD+ due to its complexity and diversity, with results presented in Table III.

The first experiment group evaluates the advantage of joint registration-change detection over alternative paradigms. Direct change detection on preregistered images achieves 88.15% IoU, representing the upper bound when perfect alignment is available. However, when applied to unregistered images, performance degrades substantially to 76.24% IoU, demonstrating the critical impact of geometric misalignment on detection accuracy. The sequential approach performing registration and change detection as independent stages recovers partial performance (87.68% IoU) but remains inferior to both the oracle preregistered case and our joint optimization. The proposed joint framework achieves 91.35% IoU, surpassing even the preregistered baseline by 3.20 percentage points. This improvement validates our hypothesis that joint optimization enables registration to be guided by change detection objectives, producing alignment specifically beneficial for semantic change analysis rather than generic image similarity.

The second experiment group analyzes individual component contributions through incremental addition. The baseline encoder–decoder achieves 86.79% IoU and 92.87% $F1$-score. Adversarial training improves IoU to 88.92% by providing discriminative feedback for feature refinement. Long-range attention further enhances performance to 90.21% IoU by capturing global semantic dependencies across image regions. Multiscale feature fusion contributes additional gains (90.78% IoU) through integration of multiresolution information for handling changes at varying scales. Cross-modal attention achieves optimal performance (91.35% IoU and 95.46%

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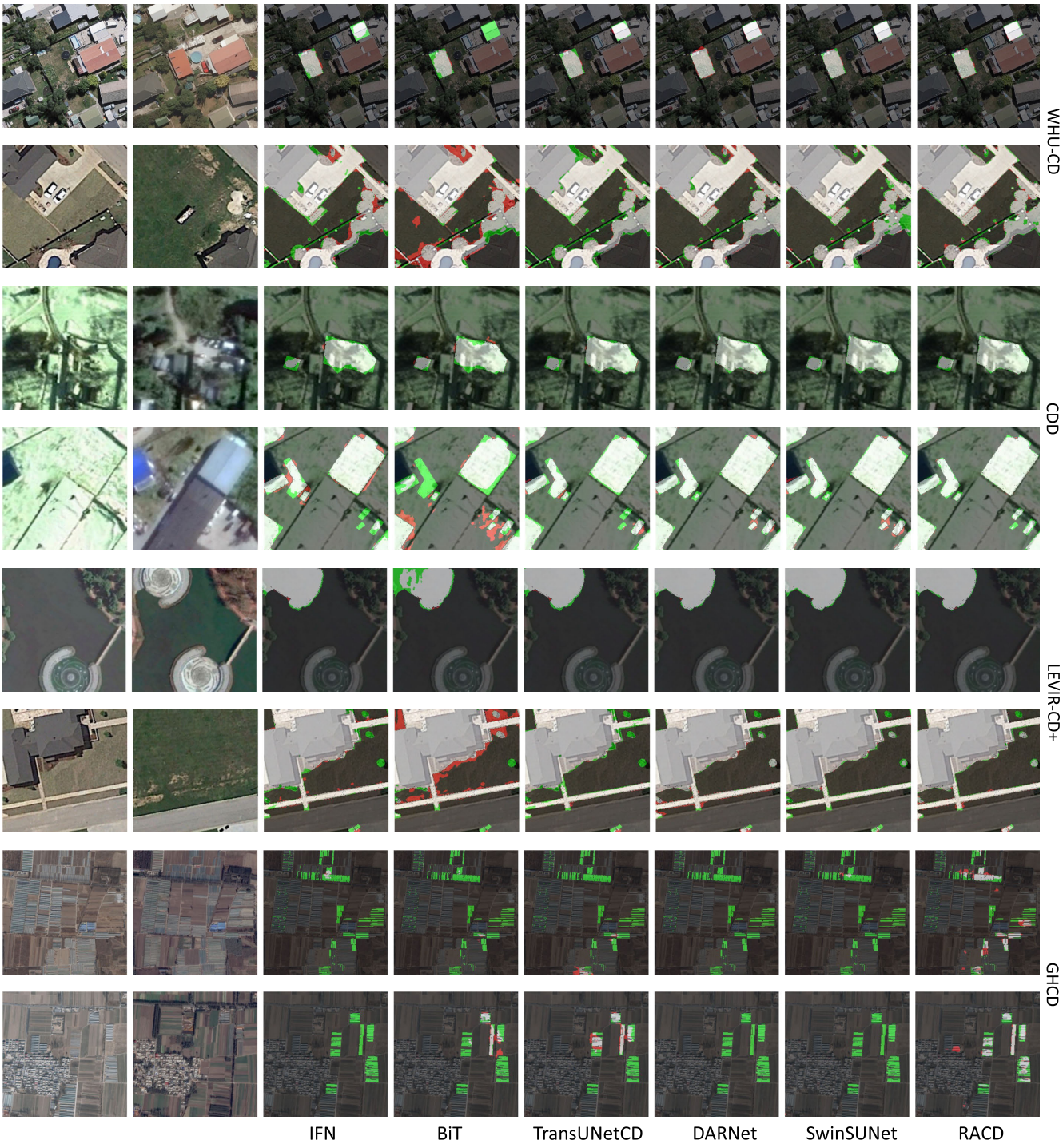


Fig. 10. Qualitative comparison of change detection results. White regions indicate correct detections, red regions denote false positives, and green regions represent missed changes.

$F1$-score) by enabling selective focus on relevant features and establishing explicit correspondences between aligned image pairs.

Hyperparameter analysis reveals that a four-layer attention configuration optimally balances performance and computational efficiency. Loss weight optimization determines the optimal ratio of 1:1:0.5 for registration, detection, and consistency losses, effectively balancing geometric alignment accuracy with change detection performance. The ablation study validates that each component provides meaningful contribution, while the joint optimization framework delivers performance gains beyond what individual improvements can achieve independently.

D. Integrated Registration and Change Detection

To evaluate the practical effectiveness of the proposed joint optimization framework, we conduct comprehensive experiments that assess the synergistic performance of registration and change detection as an integrated pipeline. This evaluation aims to demonstrate the advantages of bidirectional

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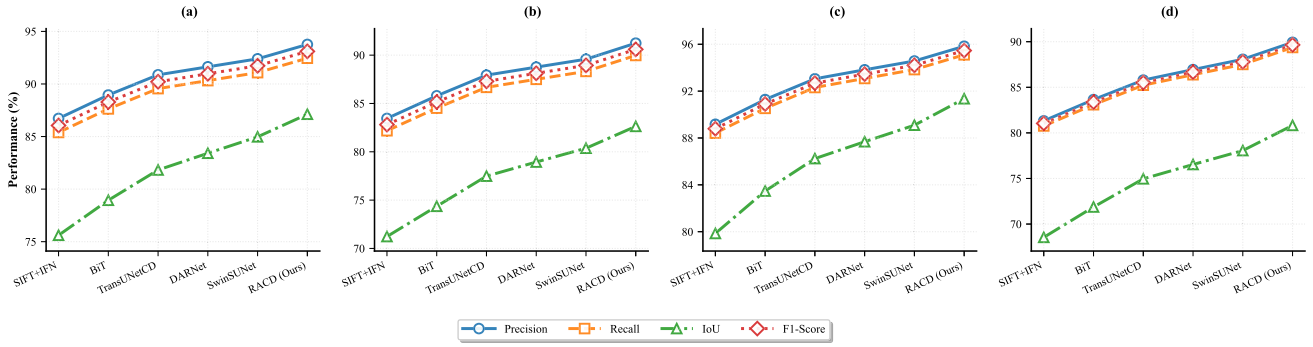


Fig. 11. Quantitative comparison of change detection methods across four datasets. (a) WHU-CD. (b) CDD. (c) LEVIR-CD+. (d) Greenhouse Dataset.



Fig. 12. Qualitative comparison of integrated registration and change detection results across four datasets.

coupling over traditional sequential approaches where registration and change detection are performed as independent stages. Experiments are conducted on WHU-CD, CDD, LEVIR-CD+, and GHCD datasets, comparing against both standalone

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TABLE IV
INTEGRATED REGISTRATION AND CHANGE DETECTION PERFORMANCE COMPARISON

Dataset	Method	P (%)	R (%)	IoU (%)	F1-Score (%)
WHU-CD	SIFT + IFN	86.72	85.39	75.61	86.05
	BiT	88.95	87.63	78.94	88.28
	TransUNetCD	90.87	89.56	81.82	90.21
	DARNet	91.63	90.32	83.41	90.97
	SwinSUNet	92.39	91.08	84.97	91.73
	Proposed	93.76	92.45	87.12	93.10
CDD	SIFT + IFN	83.45	82.18	71.23	82.81
	BiT	85.79	84.52	74.36	85.15
	TransUNetCD	87.93	86.67	77.48	87.29
	DARNet	88.76	87.50	78.94	88.12
	SwinSUNet	89.58	88.32	80.37	88.94
	Proposed	91.24	89.97	82.63	90.60
LEVIR-CD+	SIFT + IFN	89.17	88.42	79.85	88.79
	BiT	91.28	90.53	83.47	90.90
	TransUNetCD	93.05	92.31	86.24	92.68
	DARNet	93.82	93.08	87.68	93.45
	SwinSUNet	94.57	93.84	89.09	94.20
	Proposed	95.83	95.10	91.35	95.46
GHCD	SIFT + IFN	81.32	80.76	68.54	81.04
	BiT	83.65	83.09	71.87	83.37
	TransUNetCD	85.79	85.23	74.96	85.51
	DARNet	86.94	86.38	76.53	86.66
	SwinSUNet	88.07	87.51	78.06	87.79
	Proposed	89.93	89.37	80.82	89.65

change detection methods and conventional registration-then-detection pipelines.

Fig. 12 presents a qualitative comparison of integrated results across representative examples. On WHU-CD, traditional registration methods such as SIFT often fail to eliminate nonrigid geometric deformations completely, causing incorrect spatial correspondences that propagate as false positives in subsequent change detection. The proposed method effectively reduces such errors through joint optimization, where registration parameters are refined based on change detection feedback. This enables the geometric alignment to prioritize accuracy in regions critical for semantic change analysis rather than optimizing generic similarity metrics uniformly across the image.

On CDD, the proposed method demonstrates superior capability in distinguishing genuine land cover changes from pseudo-changes induced by seasonal variations. When processing image pairs containing extensive vegetation, the method accurately identifies building changes while suppressing false detections caused by phenological variations. This discrimination capability emerges from the joint training objective that encourages the change detection module to provide supervisory signals guiding registration toward semantically meaningful alignment.

On LEVIR-CD+, the method exhibits outstanding performance in handling large-scale building changes. Compared with sequential approaches, the generated change maps feature more precise boundaries and better preservation of complex building geometries. The structural consistency between registration and detection objectives ensures that geometric alignment accuracy directly translates to improved change localization, particularly for newly constructed building complexes with intricate contours.

On GHCD, the method demonstrates strong adaptability to domain-specific detection tasks. Subtle greenhouse structural changes, including local expansion and partial demolition, are accurately identified, benefiting from the multiscale feature integration that simultaneously captures global structural patterns and local detail variations. The joint learning mechanism enables registration and detection to form complementary advantages, significantly enhancing robustness in complex agricultural scenes.

Table IV presents a quantitative comparison across all datasets. The proposed method achieves optimal performance in precision, recall, $F1$-score, and IoU consistently across all four datasets, demonstrating stability and superiority in diverse scenarios. On WHU-CD, $F1$-score and IoU reach 93.10% and 87.12%, exceeding SwinSUNet [44] by 1.37 and 2.15 percentage points, respectively. On CDD, $F1$-score and IoU of 90.60% and 82.63% surpass SwinSUNet by 1.66 and 2.26 percentage points, demonstrating effectiveness in handling seasonal interference. On LEVIR-CD+, the method achieves 95.46% $F1$-score and 91.35% IoU, outperforming SwinSUNet by 1.26 and 2.26 percentage points, highlighting superior capability for large-scale structural change detection. On GHCD, $F1$-score and IoU of 89.65% and 80.82% exceed SwinSUNet by 1.86 and 2.76 percentage points, validating adaptability to domain-specific applications.

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Compared with traditional sequential pipelines (SIFT + IFN), the proposed method achieves substantial improvements across all datasets. On LEVIR-CD+, $F1$-score and IoU improvements reach 6.67 and 11.50 percentage points, respectively. This significant advancement validates the effectiveness of joint optimization in overcoming the error accumulation inherent in sequential approaches. The bidirectional coupling mechanism enables registration to benefit from change detection supervision while change detection exploits geometrically consistent features, creating synergistic enhancement that neither task could achieve independently.

The consistent high performance across datasets with diverse characteristics—complex urban landscapes (WHU-CD), seasonal variations (CDD), large-scale building changes (LEVIR-CD+), and domain-specific targets (GHCD)—demonstrates the strong generalization capability of the proposed framework. These results confirm that integrating registration and change detection through joint optimization provides substantial practical advantages over conventional approaches, offering reliable technical support for high-precision remote sensing change analysis in real-world applications. While the current study focuses on multitemporal optical data, the architecture—particularly the modality-invariant feature projection and graph-based correspondence reasoning—is inherently extensible to multimodal inputs. Exploring this direction constitutes an important part of our future research.

V. Conclusion

This article proposes an end-to-end joint learning framework for remote sensing image registration and change detection that establishes bidirectional coupling between these traditionally sequential tasks. The framework integrates similarity contrast spatial mapping using contrastive learning, matchpoint displacement regression with GCNs, and change detection using CGANs with ViTs. Through a multitask joint loss function, the method achieves mutual enhancement between registration and detection tasks, effectively handling geometric distortions and multitemporal feature heterogeneity. Experimental validation on four benchmark datasets (WHU-CD, CDD, LEVIR-CD+, and GHCD) demonstrates superior performance with $F1$-scores of 93.10%, 90.60%, 95.46%, and 89.65%, respectively, consistently outperforming existing state-of-the-art methods. The proposed joint optimization paradigm provides a robust solution for operational remote sensing applications in environmental monitoring, disaster assessment, and urban planning, while opening new possibilities for future advances in multimodal Earth observation.

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